Rajiv Gandhi Institute of Petroleum Technology, Jais Amethi

Department of Computer Science and Engineering

Mid-Semester Examination 2024-25

Subject: Object-Oriented Methodologies **Course Number: CS224**

Date & Time: 28-09-2024, 03:00 PM to 05:00 PM **Total Marks:** 30

INSTRUCTIONS

I. **This paper contains a total of six questions, with the marks for each question indicated alongside it. Students are making any necessary assumptions as required for clarification.**

1. Write a C++ program that reads a positive integer entered by an interactive user and then prints all the positive divisors of that integer in decreasing order. The program should allow the user to repeat this process as many times as the user likes. Initially, the program should inform the user about how the program will behave. Then the program should prompt the user for each integer that the user wishes to enter. The program may be terminated in any of two ways. One way is to have the program halt if the user enters an integer that's negative or zero. In this case, the user should be reminded with each prompt that the program can be terminated in that way. Alternatively, after an integer has been entered and the divisors have been printed, the program can ask the user whether he/she wishes to enter another integer. In this case, when the user accidentally enters a zero or negative integer to have its divisors calculated, the program should inform the user that the input is unacceptable and should allow the user to try again. **[5]**

2. **(a)** Differentiate between Object-Oriented Programming (OOP) and Structural Programming, mentioning at least five key differences. Additionally, explain the benefits of using OOP in real-world applications with relevant examples. **[3]**

(b) Why do encapsulation and abstraction play a vital role in providing data protection and simplification for complex systems? Justify your answer with a suitable example. **[3]**

3. Define a C++ class of private square matrix 3×3 of integers using appropriate numbers of data member and member functions. Create a parametrized constructor for the dynamic allotment of memory to the square matrix. Create a member function to initialize all the elements of the square matrix with random values. Create one more member function that finds the trace (sum of diagonal elements) of a matrix and displays the trace value. Lastly, define a destructor for deallocating the memory. Create an object of square matrix class inside the main () and then access the trace member function with this object **[5]**

4. Write a C++ program that allows the user to input a number of integers from the keyboard, and then stores them in an int array. Write a function called maxint() that goes through the array, element by element, looking for the largest one. The function should take as arguments the address of the array and the number of elements in it, and return the index number of the largest element. The program should call this function and then display the largest element and its index number. **[4]**

5. Find the output of the following questions. Also, give a proper justification for your answer. If there is any compile time error, then mentation the reason of that error. **[5]**

(a)
```
#include <iostream>
using namespace std;
class Player
{
private:
int id;
static int next_id;
public:
   int getID() { return id; }
   Player()
  {  id += ++next_id * next_id++;
  } };
int Player::next_id = 1;
int main()
{
  Player p1;
  Player p2;
  Player p3;
  cout << p1.getID() << endl;
  cout << p2.getID() << endl;
  cout << p3.getID()<<endl;
  return 0;
}
```

(b)
```
#include<iostream>
using namespace std;
class div
{
int p, q, res;
void operation()
{
total = p*p*p / (q-1);
cout <<total; }
public:
div (int a, int b)
{
p=a; q=b;
}  };
int main ()
{
div d (4,9);
d.operation();
return 0;
}
```

(c)
```
#include<iostream>
using namespace std;
class sum_square
{
  int a, b, x=0, sum=0;
public:
sum_square (int p, int q)
{ a=p; b=q;
 cout << "constructor called"; }
 void sumsqaure ()
{     x= a*a +b*b;      }
~sum_square (int x)
{  sum=x;
cout<< sum;
} };
int main ()
{       sum_square S(3,4);
        S. sumsqaure();
        return 0;
}
```

6. Find the correct answer of the following questions. Also, give a valid reason of your answer.

(a) Which among the following is true? **[1]**

(I) Default constructor can't be defined by programmer.

(II) Default parameters constructor isn't equivalent to the default constructor.

(III) Default constructor can be called explicitly.

(IV) Default constructor is and always called implicitly only.

(b) The private data members can be accessed through functions declared under **[1]**

(I) private section

(II) No way to access private data members

(III) public section

(IV) protected section

(c) The correct syntax of 'for' loop is **[1]**

(I) for (i=0; i<n; i++) {----}

(II) for(i=0,j=0; i<n && j<n; i++,j++) {-----}

(III) for(i=0; i++<n;){-----}

(IV) All of the above

(d) If two classes have exactly the same data members and member function and only they differ by class name. Can copy constructor be used to initialize one class object with another class object? **[1]**

(I) Yes, possible because the data members are same

(II) Yes, possible because member's functions are same

(III) No, but possible if constructor is also same

(IV) Not, possible

(e) In OOP, what does encapsulation refer to? **[1]**

(I) Storing classes and objects

(II) Combining data and methods

(III) Creation of classes

(IV) Creation of objects

****************End of Question Paper ****************